

On the water

The steam engine (see pp.52–53) made ships driven by paddle wheels or propellers a practical alternative to sail power. Today, ships for carrying passengers or cargo, as well as warships, have engines that turn propellers underwater to push the vessels forward or backward. While the first steam-powered ships appeared in the 1830s, most ships now have diesel engines.



Model of hull of
SS Francis Smith, 1836

A NEW SCREW

Early steamships had paddle wheels, which were easily damaged by waves. In the 1830s, the Swedish inventor John Ericsson and British inventor Francis Pettit Smith developed a propeller shaped like a corkscrew. These “screw” propellers were more powerful than paddle wheels and worked better in stormy seas.



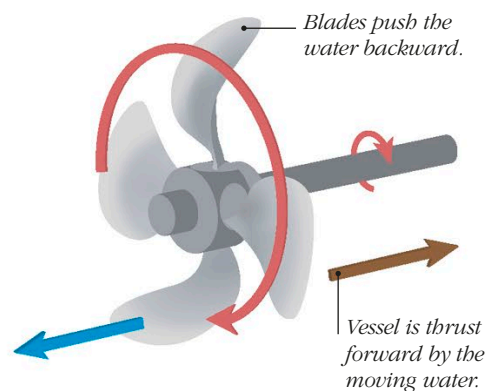
Launch of the
SS Great Britain

PROPELLING FORWARD

In 1843, the British engineer Isambard Kingdom Brunel launched his pioneering steamship, *SS Great Britain*, the most innovative passenger liner of its time. It was the first vessel to combine steam power, a screw propeller, and an iron hull. The *SS Great Britain* was the biggest ship in the world at the time, built to cruise quickly across the ocean. By the mid-19th century, propellers would replace paddle wheels on most ships.

PROPELLER IN ACTION

Propellers were first developed by the ancient Greek inventor Archimedes 2,300 years ago, but they only came into their own with the invention of steamships. A ship's propeller, or “screw,” works by moving water as it spins. The angled blades push the water backward, which pushes the boat forward.



NUCLEAR VESSEL

Ships venturing into the frozen waters of the Arctic Ocean need to be strong enough to break through thick ice sheets and to run for a long time. Russian “icebreakers” are powered by nuclear reactors, which allow them to be at sea for years without refueling.